# UMANG GUPTA

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Senior Software Engineer
Frontend Architecture & Product Engineering

Senior Software Engineer with **9+ years of experience, including 6+ years at Uber**, building scalable customer-facing web applications and frontend platforms. Strong expertise in **React, TypeScript, JavaScript, GraphQL, frontend architecture, API integration, web performance, and testing**. Proven experience leading complex frontend initiatives, designing micro-frontend and component architectures, mentoring engineers, and delivering high-quality SaaS products at scale. Experienced in collaborating across product and engineering teams to translate complex workflows into intuitive, performant web experiences.

## EXPERIENCE

### UBER — Senior Software Engineer — Aug 2020 – Present

Remote, India

- **Architected and led a frontend platform for managing risk decision trees** using React, TypeScript, ReactFlow, and GraphQL, supporting creation, visualization, comparison, versioning, and governance across **Risk, Safety, and AdTech**.
- **Designed and built** the frontend for a real-time Circuit Breaker system that detects anomalous rule behavior and **enables automated pausing of risky rules**, protecting **4,100+ live fraud rules** on a platform processing **15B+ decisions/day (250K+ QPS peak)** and preventing approximately **$2M/year in losses**.
- **Architected and built the frontend for a GenAI Copilot**, a reusable React-based chat UI library with layered architecture, contextual triggers, side-panel integration, and reusable interaction patterns; estimated to save **2–3 months of development across teams**.
- **Improved frontend performance and loading experience** through bundle splitting and server-side authorization, reducing the main JS bundle by **55.7%**, network requests by **32.4%**, and transferred resources by **32.1%**, while improving page load by **14%**.
- **Built decision-tree debugging tools for tracing fraud decisions from root to leaf**, implementing an **O(h) traversal algorithm**; reduced investigation/navigation time by **~70%** and enabled investigators to jump directly from a fraud event to the underlying decision logic.
- **Led frontend architecture and delivery for Uber's risk-rule management platform**, integrating the first gRPC service and influencing a backend cache redesign that reduced latency from **~24s to ~600ms (40×)**; improved or deprecated **339+ low-precision rules**, reducing erroneous risk actions affecting ~16.6K legitimate users and ~70K trips/orders per week, contributing to **$127.6M in incremental gross bookings**.
- **Led cross-team frontend architecture, API-contract design, and UX decisions** for Uber's internal web platform; designed micro-frontend boundaries within a monorepo, enabling multiple teams to independently develop and integrate frontend experiences; mentored engineers, created onboarding materials, and contributed to **50+ technical interviews**.
- **Led an 8-engineer frontend modernization effort** migrating **~1,400 tests across ~500 files** from Enzyme to React Testing Library as part of the React modernization effort; resolved ~200 lint issues and established standardized testing patterns for maintainable React applications.
- **Led frontend architecture and API integration for six critical risk-management workflows**, migrating them from legacy services to a gRPC-based backend using traffic shadowing and feature flags; achieved **zero user impact during migration and >30% latency improvement**.
- **Architected a standardized AI context-engineering framework with 20+ reusable, version-controlled skills** compatible with multiple LLM tools and MCPs, enabling engineers to share and maintain reusable AI workflows and development knowledge directly within the codebase.

### INDMONEY — Founding Software Engineer — Mar 2019 – Jul 2020

Gurugram, India

- Joined as the **first web engineer** and built the customer-facing web platform from 0→1, defining the React/TypeScript architecture, coding standards, and testing practices; subsequently hired **2 web engineers** and mentored **2 interns**.

- Scaled the frontend platform to approximately **500K users and 200K MAU**, supporting ~30 product and customer workflows across the investment platform.
- Designed a **configuration-driven onboarding platform** supporting multiple customer types and powering onboarding for approximately **500K users**.
- Owned customer-facing mutual fund and PMS transaction/payment experiences integrated with Razorpay, supporting approximately **₹80million/month** in transaction value; established frontend testing practices reaching **90% coverage across ~1,800 tests** and improved application performance by ~70%.

**INFOGAIN — Software Engineer** **Jul 2017 – Mar 2019**

Noida, India

- Worked on **Mitchell Connect Estimating**, a customer-facing automotive insurance/repair estimation platform used by approximately **1 lakh users** across insurers, repair shops, and estimators.
- Owned the **Paintless Dent Repair (PDR)** estimation module end-to-end, from frontend architecture and implementation through testing and production delivery.
- Introduced **React into an existing AngularJS application**, building reusable frontend components across multiple modules; collaborated directly with US-based clients and contributed ~400 automated tests.

## EARLIER EXPERIENCE

**ITC Limited: Research Intern** **Jun 2016 – Jul 2016**

Built a **Python/OpenCV computer-vision attendance system** using CCTV input for employee face detection, recognition, registration, and automated attendance management with SQLite, PyQt, and scikit-learn.

**Indian Space Research Org: Research Intern** **Jun 2015 – Jul 2015**

Built a **Python-based satellite imagery visualization system** using NumPy and Matplotlib for vegetation and phenology analysis of TIFF/TIF remote-sensing imagery.

## EDUCATION

**B.Tech, Computer Science & Engineering** **2013 - 2017**

ABES Engineering College, India

## TECHNICAL SKILLS

**Frontend:** React, TypeScript, JavaScript, HTML/CSS, Base Web, GraphQL, Apollo, Component Architecture

**Architecture & APIs:** Frontend Architecture, Micro-Frontend Architecture, Monorepo Architecture, System Design, API Design, REST, GraphQL, gRPC, HTTP/2

**Performance & Quality:** Web Performance, Bundle Optimization, React Testing Library, Jest, Playwright, Storybook

**Backend & Platform:** Node.js, Webpack, Developer Tooling

**AI:** GenAI/LLM Applications, MCP, AI-Assisted Development